HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | LinkedIn | **Stamp 1G** — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Designed and documented Single Line Diagrams (SLDs) and wiring schematics for low-voltage (LV) switchgear panels, ensuring adherence to **IEC 61439** standards, which reduced design iteration cycles by **15%**.
- Executed electrical calculations, including load calculations and short-circuit analysis, to determine appropriate **busbar sizing** and component selection, improving system fault tolerance by **20%**.
- Selected and specified critical components such as **circuit breakers**, protection relays, and contactors, coordinating with mechanical design teams to integrate these into panel layouts and achieve design freeze within project timelines.
- Developed comprehensive technical documentation, including Bill of Materials (BOM) and test procedures, supporting the factory acceptance tests (FAT) and enhancing manufacturing clarity, leading to a **10%** reduction in production errors.
- Automated repetitive tasks in the design process using **Python scripting**, reducing manual effort in generating routine schematics and BOMs by **25%**, allowing more focus on complex design challenges.
- Resolved production and testing issues by troubleshooting electrical design aspects and optimizing solutions for cost, safety, and performance, contributing to a **5%** improvement in overall product efficiency.

SKILLS

Electrical Design – LV Switchgear Design, Single Line Diagrams (SLDs), Wiring Diagrams, Schematics, Panel Layouts, Component Selection, Circuit Breakers, Busbars, Protection Relays, Contactors

Standards & Compliance – IEC 60439, IEC 61439, NSAI IS10101:2025

Electrical Calculations – Load Calculations, Short-Circuit Analysis, Thermal Analysis, Busbar Sizing

Design Tools – AutoCAD Electrical, EPLAN, (or similar)

Software Development & Automation – Python, Java, Spring Boot, SQL, REST APIs, CI/CD, Jenkins, Git

Cloud & Infrastructure – AWS, Docker, Linux

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Data Modelling

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Automated LV Switchgear Design Package Generator (HCL Technologies)

- Developed a Python-based application to automate the generation of design packages for LV switchgear, integrating with existing **AutoCAD Electrical** templates and **PostgreSQL** databases.

- Enabled automated creation of Single Line Diagrams (SLDs), wiring schematics, and Bill of Materials (BOM) based on input parameters for load and component specifications, reducing manual design time by **40%**.
- Implemented validation checks against **IEC 61439** standards within the automation script, ensuring design compliance and minimizing errors before review, leading to a **25%** faster approval cycle.

Power BI Financial Operations Dashboard Enhancement

- Enhanced a **Power BI** dashboard for a financial services client, integrating real-time transaction data to provide improved visibility into payment flows and operational alerts.
- Developed complex **DAX** measures and optimized data models to support advanced analytics on reconciliation workflows, reducing report loading times by **30%**.
- Collaborated with business analysts to refine data visualizations and KPIs, ensuring clear communication of system status and performance metrics to stakeholders.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)